

ENGINEERING SPECIFICATIONS

Document Name: IO list for KAS Siemens Double Circuit		
Document No: ESP-23-24-008	Revision: R01	Date: 29/08/2023
Prepared BY: SAM		Approved By: CGH

Analog Input/Output	POL688.10	X1	Suction Pressure Comp 1
		X2	Discharge Pressure Comp-1 (4-20 mA)
		X3	Oil Feed Pressure Comp 1(4-20 mA)
		X4	Evaporator Water In Temperature
		X5	Evaporator Water Out Temperature
		X6	SPARE
		X7	Suction Temperature Comp 1
		X8	Discharge Temperature Comp 1
		X9	Liquid Line Temperature
		X10	Ambient Temp
		X11	Suction Temperature Comp 2
	POL985	X1	VFD command Comp 1 (Analog output 0-10 Vdc)
		X2	Economiser EXV Comp1(Analog Output 0-10 Vdc)
		X3	Desuperheater Out Temperature
		X4	EXV Command comp 1(Analog Output 0-10 Vdc)
		X5	Oil Cooler EXV Command Comp 1(Analog Output 0-10 Vdc)
		X6	Evaporator Flow Meter
		X7	Suction Pressure Comp 2
		X8	Economiser Out Pressure Comp 1
		B1	Economiser Out Temperature Comp 1
		B2	Oil Temperature Comp 1
		B3	Desuperheater In Temperature
	POL985	X1	Discharge Pressure Comp 2 (4-20 mA)
		X2	Oil Feed Pressure Comp 2 (4-20 mA)
		X3	SPARE
		X4	EXV Command Comp 2
		X5	Liquid line temp Comp 2

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	X6	Economiser EXV Comp 2 (Analog Output 0-10 Vdc)
	X7	Economiser Out Pressure Comp 1 (4-20 mA)
	X8	Oil Cooler EXV Command Comp 2
	B1	Discharge Temperature Comp 2
	B2	Economiser Out Temperature Comp 2
	B3	Oil Temperature Comp 2

Digital Input	POL688	D1	Evaporator Flow switch
		D2	Condenser Flow Switch
		DU1	Comp 1 ON/OFF Switch (24V AC/DC)
		DU2	Voltage Protection Device/VFD Fault (24V AC/DC)
		DL1	Transition Feedback Comp 1 (AC 115...230 V)
		DL2	Transition Feedback Comp 2(AC 115...230 V)
	POL985	D1	SPARE
		D2	External Emergency Stop
		D3	Main On Off Switch
		DL1	Winding Protection Comp 1 (AC 115...230 V)
		DL2	Winding Protection Comp 2 (AC 115...230 V)
	POL985	D1	Comp 2 ON/OFF Switch
		D2	Voltage Protection Device/VFD Fault Comp 2
		D3	SPARE
		DL1	LP Cutout (AC 115...230 V)
		DL2	HP Cutout (AC 115...230 V)

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Digital Output	POL688	Q1	Compressor (Delta) Comp/VFD Latch Comp 2
		Q2	Compressor (Delta) Comp/VFD Latch Comp 1
		Q3	Compressor (Main) Comp 1
		Q4	Compressor (Main) Comp 2
		Q5	Comp 1 Fan 1 Motor Contactor Coil (Cond bank 1)
		Q6	Liquid Injection Solenoid Comp 1
		Q7	Fast Unloading Coil Comp 1
		Q8	Fast Unloading Coil Comp 2
		DO1	Loading Coil (V1) Potential Free Comp 1 (Triac Output-AC 24 V...230 V)
		DO2	Unloading Coil (V2) Potential Free Comp 1
	POL985	Q1	SPARE
		Q2	Liquid Injection Solenoid Comp 2
		Q3	Comp 1 Fan 2 Motor Contactor Coil
		Q4	Evaporator Pump Potential Free(on/off)
		Q5	Comp 1 Alarm Potential Free
		Q6	Comp 2 Alarm Potential Free
		Q7	Comp 1 Fan 2/3 Motor Contactor Coil (Cond bank 2)
		Q8	Comp 1 Fan 4/5 Motor Contactor Coil (Cond bank 3)
		DO1	Oil Heater Relay Comp 1
		DO2	Oil Heater Relay Comp 2
	POL985	Q1	Comp 1 Fan 6/7 Motor Contactor Coil (Cond bank 4)
		Q2	Comp 1 Fan 8/9 Motor Contactor Coil (Cond bank 5)
		Q3	Comp 1 Fan 10/11 Motor Contactor Coil (Cond bank 6)
		Q4	Comp 2 Fan 2/3 Motor Contactor Coil (Cond bank 2)
		Q5	Comp 2 Fan 4/5 Motor Contactor Coil (Cond bank 3)

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	Q6	Comp 2 Fan 6/7 Motor Contactor Coil (Cond bank 4)
	Q7	Comp 2 Fan 8/9 Motor Contactor Coil (Cond bank 5)
	Q8	Comp 2 Fan 10/11 Motor Contactor Coil (Cond bank 6)
	DO1	Loading Coil (V1) Potential Free Comp 2
	DO2	Unloading Coil (V2) Potential Free Comp 2

Controller T-IP Ethernet	PVR 7 INCH Communication Port
POL688	Main Board
POL985	Expansion Board
Controller T-HI Port	Keypad Display